The thermal-purity recursion in $2+1$ dimensions

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Abstract

For a frustration-free Hamiltonian with exact ground energy zero, the natural recursion variable is the partition-function excess $R = Z - 1$. Temporal doubling squares this excess from above. Two scalar spatial gluing inequalities then give $R' \le (1+R^2)^4 - 1 \le 5R^2$ for $R \le 0.01$. A single finite-volume seed yields a uniform gap on dyadic tori; spatial moment interpolation extends it to intermediate periods. We also retain the general three-purity cube recursion, derive an error-tolerant gluing update, and give an exact local obstruction to the standard time-reversal reflection-positive construction for AKLT. The scalar spatial gluing inequalities remain unproved here for AKLT. For a separate exact projection filter, we certify quadratic spatial concentration at transverse width two and depth one despite nonreal transfer eigenvalues. Extending this estimate as width and depth grow remains open.

1 Quadratic Excess Recursion for Frustration-Free Systems

Suppose $H_{n,m} \ge 0$ has at least one zero-energy ground state for each periodic rectangle considered. Set

$$Z(n,m,\beta) = \operatorname{Tr} e^{-\beta H_{n,m}}, \qquad R(n,m,\beta) = Z(n,m,\beta) - 1.$$

One zero-energy eigenvector contributes exactly one to Z. If the remaining energies, including any additional zeros, are E_k, then

$$R(n,m,2\beta) = \sum_{k\ge 1} e^{-2\beta E_k} \le \left(\sum_{k\ge 1} e^{-\beta E_k}\right)^2 = R(n,m,\beta)^2. \tag{1}$$

This does not assume uniqueness. A bound $R < 1$ subsequently proves uniqueness, since any additional zero-energy state contributes one.

The spatial input is only the pair of scalar inequalities

$$Z(2n,m,b) \le Z(n,m,b)^2, \tag{2}$$

$$Z(2n,2m,b) \le Z(2n,m,b)^2. \tag{3}$$

For one recursion step these are needed at $b = 2\beta$. An exact self-adjoint spatial-transfer representation implies them, but is a stronger assumption: no spatial squaring lemma is used here.

Theorem 1 (Frustration-free $2+1$D recursion)**.** *If* (2)–(3) *hold at* $b = 2\beta$*, then*

$$\boxed{R(2n,2m,2\beta) \le \left[1 + R(n,m,\beta)^2\right]^4 - 1.} \tag{4}$$

In particular, for $0 \le R \le 0.01$*,*

$$\boxed{R' \le 4R^2 + 6R^4 + 4R^6 + R^8 \le 5R^2.} \tag{5}$$

Proof. First double imaginary time using (1); then double the two spatial periods using (2)–(3):

$$Z(2n, 2m, 2\beta) \le Z(n, m, 2\beta)^4 \le \big[1 + R(n, m, \beta)^2\big]^4.$$

Subtract one. For $R \le 0.01$, the coefficient of R^2 is at most $4+6(0.01)^2+4(0.01)^4+(0.01)^6 < 5$. □

The order matters for the bound: temporal doubling first gives the leading coefficient 4. Spatial doubling first would give the weaker $R' \le [(1 + R)^4 - 1]^2$, with leading coefficient 16. In d spatial dimensions the same argument gives $R' \le (1 + R^2)^{2^d} - 1$, with leading coefficient 2^d.

One finite seed. Assume the gluing inequalities at every subsequent dyadic step, and certify only

$$Z(n_0, m_0, \beta_0) \le 1.01. \tag{6}$$

With $(n_j, m_j, \beta_j) = 2^j(n_0, m_0, \beta_0)$, induction gives

$$R(n_j, m_j, \beta_j) \le r_j := \frac{1}{5}\, 20^{-2^j}, \qquad r_{j+1} = 5r_j^2. \tag{7}$$

The ground state is unique on each such torus and $e^{-\beta_j \Delta_{n_j,m_j}} \le R(n_j, m_j, \beta_j)$. Thus $\Delta_{n_j,m_j} > (\log 20)/\beta_0$, uniformly in j. There is no need to reconstruct the gap from a normalized purity.

If a finite seed Hamiltonian has a unique zero-energy ground state, Hilbert-space dimension D, and a certified finite-volume gap $\gamma_* > 0$, then

$$R(n_0, m_0, \beta) \le (D - 1)e^{-\beta\gamma_*}.$$

Consequently, $\beta_0 = \log[100(D - 1)]/\gamma_*$ suffices for (6). Under the spatial assumptions, this yields the explicit, possibly small, lower bound

$$\Delta_{n_j,m_j} > \frac{\gamma_* \log 20}{\log[100(D - 1)]}. \tag{8}$$

Existence of a finite-volume seed is therefore automatic from finite dimensionality and uniqueness. The substantive uniform input is the spatial gluing property, not the existence of three low-temperature purity certificates.

Intermediate periods require an additional input. If the spatial moment representations introduced below also hold, then for even $n_j \le L_x \le 2n_j$ and $m_j \le L_y \le 2m_j$,

$$Z(L_x, L_y, \beta_j) \le Z(n_j, m_j, \beta_j)^{L_x L_y/(n_j m_j)} \le (1 + r_j)^4.$$

Since $(1 + r)^4 - 1 < 5r$ for $0 < r \le 0.01$,

$$e^{-\beta_j \Delta_{L_x,L_y}} \le R(L_x, L_y, \beta_j) < 5r_j = 20^{-2^j}.$$

The same gap bound follows on these windows. In particular, equal seed periods cover all sufficiently large even square tori. Scalar doubling inequalities alone establish the dyadic result, not this interpolation.

Errors in spatial gluing. The exact inequalities can be relaxed. Suppose at $b = 2\beta_j$,

$$Z(2n_j, m_j, b) \le e^{\eta_{x,j}} Z(n_j, m_j, b)^2,$$
$$Z(2n_j, 2m_j, b) \le e^{\eta_{y,j}} Z(2n_j, m_j, b)^2,$$

with nonnegative errors. Put $\eta_j = 2\eta_{x,j} + \eta_{y,j}$. The same proof gives the rigorous perturbed update

$$\boxed{R_{j+1} \le e^{\eta_j}(1 + R_j^2)^4 - 1.} \tag{9}$$

For $R_j, \eta_j \le 0.01$, this is at most $5R_j^2 + 2\eta_j$. A constant nonzero gluing error is therefore insufficient for this iteration to prove a gap; it creates an error floor. An estimate $\eta_j \le cR_j^2$ instead retains a quadratic recursion with coefficient $5 + 2c$. PEPS boundary estimates would need an explicit bridge to these thermal partition-function errors before they could be used as this input.

2 Directional Purities and Spatial Transfer Operators

Let $H_{n,m}$ be a finite-dimensional Hamiltonian on a periodic two-dimensional lattice, and put

$$Z(n, m, \beta) = \operatorname{Tr} e^{-\beta H_{n,m}}, \qquad \beta > 0.$$

The integers n, m count spatial slices or fixed unit cells, not necessarily individual spins. We use even periods throughout. Write $v = (n, m, \beta)$, with commuting doubling operations

$$D_x v = (2n, m, \beta), \quad D_y v = (n, 2m, \beta), \quad D_t v = (n, m, 2\beta).$$

Define

$$P_x(v) = \frac{Z(2n, m, \beta)}{Z(n, m, \beta)^2}, \qquad P_y(v) = \frac{Z(n, 2m, \beta)}{Z(n, m, \beta)^2}, \qquad P_t(v) = \frac{Z(n, m, 2\beta)}{Z(n, m, \beta)^2}. \tag{10}$$

Here t denotes imaginary time. The temporal ratio is always the purity of $\rho_v = e^{-\beta H_{n,m}}/Z(v)$.

Spatial-transfer hypothesis. For every transverse period and inverse temperature used below, assume there exist self-adjoint operators $T_x(m, \beta)$ and $T_y(n, \beta)$, independent of their respective longitudinal periods, such that

$$Z(n, m, \beta) = \operatorname{tr} T_x(m, \beta)^n = \operatorname{tr} T_y(n, \beta)^m. \tag{11}$$

Assume the even powers in these identities are positive trace-class operators. No uniform bound on the transfer-space dimension is needed. The operators for different directions need not act on the same space or commute. The identities must be exact; an unbounded truncation error cannot be inserted into the argument.

For even n, m, the trace-one operators

$$\sigma_x(v) = \frac{T_x(m, \beta)^n}{Z(v)}, \qquad \sigma_y(v) = \frac{T_y(n, \beta)^m}{Z(v)}, \qquad \sigma_t(v) = \rho_v$$

are positive, and $P_a(v) = \operatorname{tr} \sigma_a(v)^2$. Consequently,

$$0 < P_a(v) \le 1, \qquad \sigma_a(D_a v) = \frac{\sigma_a(v)^2}{P_a(v)}, \qquad a \in \{x, y, t\}. \tag{12}$$

An exact tensor-network representation alone does not imply (11): spatial reflection, positivity, and the identification of the same transfer operator at different lengths must be checked.

3 Squaring Lemma and Spacetime Cube Coupling

For any positive trace-one operator with eigenvalues p_i, write $P = \sum_i p_i^2$. Its normalized square has purity P', with

$$1 - P' = \frac{2\sum_{i<j} p_i^2 p_j^2}{P^2} \le \frac{(1-P)^2}{2P^2}.$$

Thus, with

$$f(z) = \frac{z^2}{2(1-z)^2}, \qquad 0 \le z < 1, \tag{13}$$

we have $1 - P_a(D_a v) \le f(1 - P_a(v))$.

Every pair of directions also satisfies the exact identity

$$P_a(v)^2 P_b(D_a v) = P_b(v)^2 P_a(D_b v). \tag{14}$$

This follows by cancellation of partition functions and requires no transfer assumption. The same cancellation on the three-dimensional cube gives, for any permutation (a, b, c) of (x, y, t),

$$Q(v) := \frac{Z(D_x D_y D_t v)}{Z(v)^8} = P_a(v)^4 P_b(D_a v)^2 P_c(D_b D_a v). \tag{15}$$

The exponents $4, 2, 1$ count how often the earlier ratios enter the successive squarings. In particular, since every factor is at most one,

$$P_c(D_a D_b v) \ge Q(v). \tag{16}$$

The cube identity supplies the coupling; positivity supplies this inequality.

4 Staggered Three-Component Scale-Doubling Iteration

At the base point v, assume

$$P_x(v) \ge 1 - \varepsilon_x, \qquad P_y(D_x v) \ge 1 - \varepsilon_y, \qquad P_t(D_y D_x v) \ge 1 - \varepsilon_t. \tag{17}$$

These are the three edges of the path x, y, t through one cube. Put $v' = D_x D_y D_t v$. The next-scale quantities occupy the same staggered positions:

$$P_x(v'), \qquad P_y(D_x v'), \qquad P_t(D_y D_x v'). \tag{18}$$

In coordinates, the input and output are

	input	output
x	$P_x(n, m, \beta)$	$P_x(2n, 2m, 2\beta)$
y	$P_y(2n, m, \beta)$	$P_y(4n, 2m, 2\beta)$
t	$P_t(2n, 2m, \beta)$	$P_t(4n, 4m, 2\beta)$.

Theorem 2 (Three-component update)**.** *Define successively*

$$u_x = (1-\varepsilon_x)^4(1-\varepsilon_y)^2(1-\varepsilon_t), \qquad \delta_x = f(1-u_x), \tag{19}$$

$$u_y = (1-\varepsilon_y)^4(1-\varepsilon_t)^2(1-\delta_x), \qquad \delta_y = f(1-u_y), \tag{20}$$

$$u_t = (1-\varepsilon_t)^4(1-\delta_x)^2(1-\delta_y), \qquad \delta_t = f(1-u_t). \tag{21}$$

Provided the input defects lie in $[0, 1)$ *and* $\delta_x, \delta_y < 1$*, the output defects in* (18) *are bounded by* $\delta_x, \delta_y, \delta_t$*, respectively.*

Proof. First use the cube at v. Its path x, y, t gives $Q(v) \geq u_x$. Reorder the path to y, t, x; by (16), $P_x(D_y D_t v) \geq u_x$. Squaring in x yields $P_x(v') \geq 1 - \delta_x$.

Next use the cube based at $D_x v$. Its path y, t, x uses two old bounds and the just-proved bound:

$$Q(D_x v) = P_y(D_x v)^4 P_t(D_y D_x v)^2 P_x(v') \geq u_y.$$

Reordering to x, t, y gives $P_y(D_x^2 D_t v) \geq u_y$. Squaring in y yields $P_y(D_x v') \geq 1 - \delta_y$.

Finally use the cube based at $D_x D_y v$. Its path t, x, y gives

$$Q(D_x D_y v) = P_t(D_x D_y v)^4 P_x(v')^2 P_y(D_x v') \geq u_t.$$

Reordering to x, y, t gives $P_t(D_x^2 D_y^2 v) \geq u_t$. Squaring in t yields $P_t(D_x D_y v') \geq 1 - \delta_t$. □

Theorem 3 (Closed scalar recursion)**.** *If the three input defects are at most $\varepsilon \leq 10^{-3}$, then all three output defects are at most*

$$g(\varepsilon) = f\big(1 - (1-\varepsilon)^7\big) = \frac{1}{2}\big[(1-\varepsilon)^{-7} - 1\big]^2 \leq 25\varepsilon^2. \tag{22}$$

Proof. Bernoulli's inequality and monotonicity of f give

$$g(\varepsilon) \leq f(7\varepsilon) = \frac{49\varepsilon^2}{2(1-7\varepsilon)^2} \leq 25\varepsilon^2 \leq \varepsilon.$$

The numerical inequality is exact: at $\varepsilon = 1/1000$, $49{,}000{,}000 < 50 \cdot 993^2 = 49{,}302{,}450$. The first update has $u_x \geq (1-\varepsilon)^7$, so $\delta_x \leq g(\varepsilon) \leq \varepsilon$. This implies $u_y \geq (1-\varepsilon)^7$, hence the same bound on δ_y; it then implies the corresponding bound on δ_t. Thus the recursion closes. □

5 Uniform Spectral Gap from Certified Seeds

Suppose three finite-scale certificates establish

$$P_x(n_0, m_0, \beta_0), \quad P_y(2n_0, m_0, \beta_0), \quad P_t(2n_0, 2m_0, \beta_0) \ \geq 0.999. \tag{23}$$

Set

$$n_j = 2^j n_0, \quad m_j = 2^j m_0, \quad \beta_j = 2^j \beta_0, \qquad \varepsilon_j = \frac{1}{25} 40^{-2^j}. \tag{24}$$

Then $\varepsilon_0 = 10^{-3}$ and $\varepsilon_{j+1} = 25\varepsilon_j^2$, so the same three staggered purities at scale j are at least $1 - \varepsilon_j$.

We can also cover intermediate even periods. Choose even L_x, L_y such that, for the same j,

$$n_j \leq L_x \leq 2n_j, \qquad m_j \leq L_y \leq 2m_j. \tag{25}$$

For fixed transverse period and temperature, (11) expresses Z as an even spectral moment. Monotonicity of the ℓ^p norms, applied successively in x and y, gives

$$\begin{aligned} Z(L_x, L_y, \beta_j) &\leq Z(n_j, m_j, \beta_j)^{L_x L_y/(n_j m_j)}, \\ Z(L_x, L_y, 2\beta_j) &\geq Z(2n_j, 2m_j, 2\beta_j)^{L_x L_y/(4 n_j m_j)}. \end{aligned}$$

With $\alpha = L_x L_y/(4 n_j m_j) \in [1/4, 1]$, this implies

$$\begin{aligned} P_t(L_x, L_y, \beta_j) &\geq Q(n_j, m_j, \beta_j)^\alpha \\ &\geq (1 - \varepsilon_j)^{7\alpha} \geq 1 - 7\varepsilon_j > \tfrac{1}{2}. \end{aligned} \tag{26}$$

If $w_0 \geq w_1 \geq \cdots$ are the physical Gibbs weights, purity is at most w_0. Thus the ground state is unique and

$$e^{-\beta_j \Delta_{L_x,L_y}} = \frac{w_1}{w_0} \leq \frac{7\varepsilon_j}{1-7\varepsilon_j} < 40^{-2^j}.$$

Consequently,

$$\boxed{\Delta_{L_x,L_y} > \frac{\log 40}{\beta_0}.} \tag{27}$$

This holds on the union of windows (25), conditional on (11) and (23). In particular, taking $n_0 = m_0$ covers every even square torus of side at least n_0. It does not assert a bound for all aspect ratios or for open boundaries. For a common seed defect $0 < \varepsilon_0 \leq 10^{-3}$, the same argument replaces 40 by $(25\varepsilon_0)^{-1}$.

6 Application and Obstructions in the Hexagonal AKLT Model

For the spin-3/2 hexagonal AKLT Hamiltonian in projector normalization,

$$H_{n,m}^{\mathrm{AKLT}} = \sum_{\langle u,v \rangle} P_{uv}^{(S_{\mathrm{tot}}=3)},$$

choose fixed unit cells and two compatible periodic slice directions. The frustration-free refinement requires:

1. Prove the two scalar gluing inequalities (2)–(3) on the required scales, or control their errors through (9). Self-adjoint transfers are one sufficient construction.

2. Bound one finite-volume partition-function excess, or use a certified finite-volume gap and dimension as in (8). Spatial moment interpolation is a separate requirement if all intermediate periods are desired.

No AKLT scalar gluing proof is supplied here. The result is therefore a proved conditional gap criterion, not an AKLT gap proof.

An exact obstruction to the standard reflection argument. This spatial input cannot simply be borrowed from the Heisenberg interaction. Consider one spin-3/2 AKLT bond $h = P^{(3)}$ and the quadrupole

$$Q = (S^z)^2 - \frac{5}{4} I = \mathrm{diag}(1, -1, -1, 1).$$

Let $\theta(A) = U\overline{A}U^\dagger$, where $U = e^{-i\pi S^y}$, be the usual spin time-reversal action used with a spatial reflection. Then $\theta(Q) = Q$. Reflection positivity for this bond would require $\mathrm{Tr}[e^{-th}(A \otimes \theta(A))] \geq 0$ for every A. But $\mathrm{Tr}\, Q = 0$, $e^{-tP^{(3)}} = I + (e^{-t} - 1)P^{(3)}$, and

$$\mathrm{Tr}[P^{(3)}(Q \otimes Q)] = \frac{7}{5}. \tag{28}$$

Hence for every $t > 0$,

$$\boxed{\mathrm{Tr}[e^{-tP^{(3)}}(Q \otimes \theta(Q))] = \frac{7}{5}(e^{-t} - 1) < 0.} \tag{29}$$

For a direct check of (28), the diagonal expectations of $Q \otimes Q$ in $|3, M\rangle$, $M = -3, -2, \ldots, 3$, are $1, -1, 1/5, 1, 1/5, -1, 1$; their sum is $7/5$. Equivalently, with $X = \mathbf{S}_1 \cdot \mathbf{S}_2$, the projector is

$P^{(3)} = (X+15I/4)(X+11I/4)(X+3I/4)/90$. The characterization of thermal reflection positivity by cross-plane couplings is discussed in Ref. [3].

This calculation rules out this particular local reflection-positive construction. It does not disprove the weaker scalar inequalities (2)–(3), or every possible blocked transfer construction. Those statements must be tested independently.

There is also a direct spectral diagnostic for scalar gluing. On two tori with unique zero-energy ground states, write $Z(n,m,\beta) = 1 + g_{n,m}e^{-\beta\Delta_{n,m}} + \cdots$. If $\Delta_{2n,m} < \Delta_{n,m}$, then $Z(2n,m,\beta) > Z(n,m,\beta)^2$ for all sufficiently large β. Thus a verified decrease of the finite-volume gap under doubling would obstruct the exact gluing inequality at those sizes.

The PEPS boundary method offers a complementary route: relative factorization of overlapping ground-state boundary operators implies control of regional ground-space projectors and hence a bulk gap [1]. The recent AKLT preprint [2] develops boundary-response estimates using paths and loops. Such estimates may inform the model-specific analysis above, but they are not automatically thermal-purity certificates. Here the objects are the full physical partition functions, rather than the PEPS boundary Gram operators.

7 Empirical Doubling Coefficients and Diagnostics

Exact spatial gluing is sufficient but not necessary for the recursion. For the dyadic sequence define the actual coefficient

$$C_j = \frac{R(n_{j+1}, m_{j+1}, \beta_{j+1})}{R(n_j, m_j, \beta_j)^2}. \tag{30}$$

If $C_j \le C$ uniformly, with $C \ge 1$ and $CR_0 < 1$, then induction gives

$$R_j \le C^{-1}(CR_0)^{2^j}, \qquad \Delta_{n_j,m_j} \ge \frac{\log(1/(CR_0))}{\beta_0}.$$

This only needs control along the chosen sequence $\beta_j/n_j = \beta_0/n_0$, rather than at all temperatures for each fixed torus. Finitely many measured coefficients do not prove the uniform bound.

A useful diagnostic of the sufficient gluing construction is

$$G_j = \log Z(2n_j, 2m_j, 2\beta_j) - 4\log Z(n_j, m_j, 2\beta_j).$$

Taking $\eta_j = \max\{0, G_j\}$ in (9) is valid directly, and can be sharper than bounding the two directional errors separately.

7.1 First computational pilot

The companion code in `models/aklt/` uses primitive cells $A(x,y), B(x,y)$, with edges from $A(x,y)$ to $B(x,y), B(x-1,y), B(x,y-1)$ and both coordinates periodic. The initial geometries are 2×2, 4×2, and 4×4 cells, containing 8, 16, and 32 spins. These small quotients have finite-size winding effects; in particular, the eight-spin quotient has short winding cycles. They are not open hexagonal patches.

Independent full-spectrum diagonalization of the eight-spin quotient includes all $4^8 = 65536$ states through total-S^z sectors. It gives one eigenvalue within 10^{-9} of zero and

$$E_0 \simeq 1.35\times 10^{-14}, \qquad \Delta_8 \simeq 0.0920298633241,$$
$$R(2,2,64) \simeq 0.00831113960.$$

The numerical threshold for $R \le 0.01$ is $\beta \simeq 61.9932$. This provides a candidate finite seed at $\beta_0 = 64$, not an interval certificate. If the seed and uniform spatial gluing were certified, the conservative recursion above would give $(\log 20)/64 \simeq 0.0468083$. Neither the full gap claim nor uniform gluing follows from this finite calculation.

The thermal implementation evolves the MPO amplitude $e^{-\beta H/2}$, using $e^{-tP^{(3)}} = I + (e^{-t} - 1)P^{(3)}$ for each local gate. It retains the full thermal trace and reconstructs $\log Z$ from normalization factors. A forward/reverse gate sweep has second-order Trotter error; compression and the swaps needed for nonlocal gates introduce additional errors.

At $\beta = 2$, full diagonalization gives $\log Z = 5.31205522615$. The following benchmark uses time step $\delta\beta = 1/8$, cutoff 10^{-12}, and the default policy of returning moved sites after each gate:

Maximum MPO bond dimension	Computed $\log Z$	Error in $\log Z$
16	3.53951535	-1.77253988
32	3.88853480	-1.42352043
64	4.72481476	-0.58724047
128	4.97859393	-0.33346129

Even this moderate-temperature benchmark is not converged. For example, at bond dimension 64, halving the time step gives $\log Z = 4.58083645$, increasing the discrepancy. The smaller time step alone does not remove compression error. Negative gluing defects or small doubling coefficients from these compressed calculations must therefore not be treated as evidence for the desired inequalities. A separate test keeps moved sites between gates and restores their order only after each complete step. At bond dimension 128 and $\delta\beta = 1/8$, this gives $\log Z(2) = 4.91941052$; reducing repeated swaps does not resolve the benchmark discrepancy.

Machine-readable spectra, convergence grids, and diagnostics are saved under `models/aklt/results/`, with source hashes, geometry, parameters, completion flags, and `certified=false`. The smallest-system checks compare the two independent projector constructions, the exact single-bond partition function, and full-spectrum four-site evolution, including its nonlocal closing bond. The immediate computational task is to resolve partition-function differences accurately before using larger thermal calculations to assess the recursion.

8 Quadratic concentration without a self-adjoint spatial transfer

The coupled squaring mechanism of the OpenAI proof [4, 5] uses scalar properties of the spatial moments. Self-adjointness is one sufficient way to obtain those properties. Nonreal transfer eigenvalues rule out a positive metric making that transfer self-adjoint, but do not by themselves rule out the scalar inequalities needed for the recursion. We establish this distinction for one exact AKLT projection-filter slice.

8.1 An exact filter and its spatial transfer

Let E_1, E_2, E_3 be the three matchings used in the pilot, with E_1 intracell, E_2 joining $A(x,y)$ to $B(x-1,y)$, and E_3 joining $A(x,y)$ to $B(x,y-1)$. Set

$$q_e = I - P_e^{(3)}, \qquad Q_c = \prod_{e \in E_c} q_e, \qquad D = Q_3 Q_2 Q_1, \qquad F = D^\dagger D = Q_1 Q_2 Q_3 Q_2 Q_1.$$

Then $0 \le F \le I$ and $\ker(I-F) = \ker H$: equality of norms through a product of orthogonal projections requires every projection to fix the vector. Define the discrete-depth moments

$$\mathcal{Z}(n,m,t) = \operatorname{Tr} F_{n,m}^{t}, \qquad t \in \mathbb{N}.$$

Temporal squaring is exact for these moments. The converse detectability inequality gives $I - F \le 4H$ [6]. Thus a uniform estimate $F \le G + q(I-G)$, where G projects onto $\ker H$ and $q < 1$, would give $\Delta(H) \ge (1-q)/4$.

Fix transverse width $n = 2$ and depth $t = 1$, and let the longitudinal period be L. A row contains the four spins A_0, A_1, B_0, B_1. Write

$$A = Q_{1,\mathrm{row}}, \qquad B = Q_{2,\mathrm{row}} A Q_{2,\mathrm{row}}, \qquad \mathcal{R}(X)_{(a,a'),(b,b')} = X_{(a,b),(a',b')}.$$

Here each of a, b indexes two spins, hence has dimension 16. Cyclicity, $Q_1^2 = Q_1$, and direct contraction of the interrow Q_3 bonds give the 256-dimensional transfer

$$T = \mathcal{R}(A)^{\mathsf{T}} \mathcal{R}(B^{\mathsf{T}}), \qquad z_L := \mathcal{Z}(2, L, 1) = \operatorname{tr} T^{L}. \tag{31}$$

This transfer has a conjugate pair $0.4111133913 \pm 0.3495535140i$, each with multiplicity five. Exact characteristic-polynomial and root-count checks establish their nonreality. Therefore neither T nor T^2 can be made self-adjoint by a positive metric. The scalar moments nevertheless satisfy a quadratic concentration estimate.

8.2 A scalar replacement for spectral positivity

Lemma 1 (Concentration from a controlled trace remainder)**.** *Suppose $z_L = \lambda_0^L(1 + r_L)$, where $\lambda_0 > 0$, and for every even $L \ge L_0$,*

$$0 < r_L \le \tfrac{1}{10}, \qquad 0 < r_{2L} \le \kappa r_L^2, \qquad 0 < \kappa \le \tfrac{34}{100}.$$

Then the scalar defect $d_L = 1 - z_{2L}/z_L^2$ obeys

$$0 < d_L < 1, \qquad d_{2L} \le \frac{\kappa}{2}\left(\frac{11}{10}\right)^4 d_L^2 < \frac{1}{4} d_L^2. \tag{32}$$

Proof. The identity

$$d_L = \frac{2r_L + r_L^2 - r_{2L}}{(1+r_L)^2}$$

gives $d_L \ge 2r_L/(1+r_L)^2 > 0$, since $\kappa < 1$, and $d_L < 1$ since $z_{2L} > 0$. Also $d_{2L} \le 2r_{2L} \le 2\kappa r_L^2$, using $r_{4L} \ge 0$. Substitute $r_L \le (1+r_L)^2 d_L/2$ and $r_L \le 1/10$. □

Exact spectral bounds for the fixed slice. The verifier finds one simple leading eigenvalue λ_0, one real eigenvalue α of multiplicity three, 242 remaining real eigenvalues, and ten nonreal eigenvalues. Exact rational root counts certify

$$9.57 < \lambda_0 < 9.59, \qquad 3.36 < \alpha < 3.37,$$
$$|\lambda| < 0.84 \quad \text{for the remaining real modes,}$$
$$|\lambda| < 0.55 \quad \text{for every nonreal mode.}$$

All displayed decimal endpoints are exact rational bounds. No diagonalizability assumption is needed: traces of powers count eigenvalues with algebraic multiplicity.

For even $L \geq 4$, let $r_L = \sum_{i>0}(\lambda_i/\lambda_0)^L$ and $u = 55/336$. Real modes contribute nonnegatively, so

$$r_L \geq 3(\alpha/\lambda_0)^L \left[1 - \frac{10}{3}u^L\right] > 0.$$

The absolute remainder has the uniform bound

$$r_L \leq 3\left(\frac{337}{957}\right)^4 + 242\left(\frac{84}{957}\right)^4 + 10\left(\frac{55}{957}\right)^4 < 0.060605 < \frac{1}{10}.$$

Bounding the numerator at $2L$ and the denominator at L gives

$$\frac{r_{2L}}{r_L^2} \leq \frac{3 + 242(1/4)^{2L} + 10u^{2L}}{9[1 - (10/3)u^L]^2} \leq \kappa_* := \frac{3 + 242(1/4)^8 + 10u^8}{9[1 - (10/3)u^4]^2} < 0.335348 < 0.34.$$

The lemma therefore proves

$$\boxed{d_{2L} \leq \tfrac{1}{4}d_L^2 \quad \text{for every even } L \geq 4, \quad z_L = \mathcal{Z}(2, L, 1).} \tag{33}$$

This is stronger than the scalar squaring inequality $d_{2L} \leq f(d_L)$ used earlier, although the spatial transfer has nonreal eigenvalues.

Verification and remaining scope. In the physical basis change $S = \operatorname{diag}\sqrt{\binom{3}{a}}$, $a = 0, 1, 2, 3$, each bond gate becomes rational with denominator 60, so $60^8 T$ is integral in the corresponding transfer basis; exact characteristic polynomials and root counts establish the spectral bounds above and match an independent eight-spin tensor contraction. The Julia pilot in `models/aklt/` (`scripts/run.jl`, `test/runtests.jl`) checks the bond projector polynomial, the quadrupole reflection obstruction (28), and the exact and thermal-purification benchmarks.

Equation (33) holds at one fixed transverse width and one fixed depth. It does not control the mixed-width, mixed-depth tuples needed by the cube recursion. The next mathematical target is to propagate such domination estimates in both transverse size and depth, so that both spatial ratios remain in $(0, 1]$ and obey their squaring bounds throughout the staggered recursion. The cube proof then applies to the scalar ratios directly, without requiring spatial density operators. No uniform AKLT gap follows from the fixed-slice certificate alone.

References

[1] M. J. Kastoryano, A. Lucia, and D. Pérez-García, *Locality at the boundary implies gap in the bulk for 2D PEPS*, Commun. Math. Phys. **366**, 895–926 (2019), arXiv:1709.07691.

[2] M. J. Kastoryano, *Spectral Gap of the Hexagonal AKLT Model via Boundary-State Factorization*, preprint (September 23, 2026), arXiv:2609.27993.

[3] A. Jaffe and B. Janssens, *Characterization of Reflection Positivity: Majoranas and Spins*, arXiv:1506.04197.

[4] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, OpenAI Math Release.

[5] Companion note, *A diagrammatic proof of the Haldane gap via spacetime tensor networks*, `notes/thermal-mpo-gap-note.tex` (2026).

[6] A. Anshu, I. Arad, and T. Vidick, *A simple proof of the detectability lemma and spectral gap amplification*, arXiv:1602.01210.